

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years building and operating **distributed, event-driven microservices** in production. Architected a **Kafka**-based microservices backend on **Docker and Kubernetes**; built a redelivery-safe, **idempotent** two-way MongoDB↔Firestore sync on Google **Pub/Sub** that carried live traffic through a **zero-downtime** migration of 20+ applications. Owns work end-to-end — from design through deployment into production and iteration — and has shipped production backends in Node.js, Python and Go. Based in Kampala, working with distributed teams across US, Singapore and Uganda time zones since 2020. Uses AI tooling (Claude) daily for design exploration, code review and test writing. Primary stack is Node.js; keen to apply the same event-driven patterns in C#/.NET on Azure.

CORE COMPETENCIES

Event-Driven Microservices

Messaging: Kafka, Pub/Sub, NATS

Idempotency & Retry Design

Kubernetes & Docker

End-to-End Production Ownership

Zero-Downtime Migrations

Distributed Teams Across Time Zones

AI-Assisted Engineering Workflow

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, owning design, implementation, deployment and production iteration; reported directly to the CTO.
- Built a real-time, two-way MongoDB↔Firestore sync service in **Node.js** on Google **Pub/Sub**, designed for **idempotent processing, redelivery and out-of-order events** so both datastores could serve live traffic in parallel.
- Consolidated all services off AWS onto GCP (file storage S3→GCS via a Python pipeline), cutting infrastructure cost by **\$5,000/month**.
- Shipped revenue features end-to-end: Express Shipping (**+\$40 revenue per order**) and a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Ramped onto an unfamiliar Firebase/GCP stack and delivered the MongoDB→Firestore migration scripts with complex transformation logic; converted to full-time and given ownership of the whole migration.
- Wrote the Firebase SDK documentation (Node.js and Python) other engineers onboarded from; trained the Ops team on the new dashboard and backup procedures.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using **Node.js** and Web3; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built scheduled jobs querying **2M+ Firestore records** to keep customer-facing stock prices current from Morningstar APIs; extended backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an **event-driven microservices** backend for FIDA Uganda's case-management platform on **Apache Kafka, Docker and Kubernetes**, with NATS Streaming for service messaging.
- Built a **USSD service for legal-aid providers** talking to a Node.js backend over **gRPC** — reaching feature-phone users across Uganda.
- Led a team of 4 developers; delivered a paralegal database for LASPNET Uganda and a cross-platform React Native (Expo) app with Firebase push notifications.

PROJECTS

Event-Driven Microservices — FIDA Uganda

Kafka-based event-driven backend for a legal case-management platform, containerised and orchestrated on Kubernetes; services consumed by web, mobile and USSD clients.

Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming · Node.js

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore sync with idempotent, redelivery-safe message handling, running under parallel production traffic across 20+ applications for the full migration window.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Chatbot

Conversational assistant on the WhatsApp Business API integrating OpenAI and Google Vision for prescription OCR.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Languages: JavaScript / Node.js, TypeScript, Python, Go (intermediate), Rust (intermediate), SQL

Messaging & Streaming: Apache Kafka, Google Pub/Sub, NATS Streaming, gRPC

Cloud & Infra: GCP (Firestore, Cloud Functions, Pub/Sub, GCS), AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Linux

Databases: MongoDB, Firestore, SQL **Tooling:** Git, Claude and AI-assisted development workflow **Spoken:** English, Kiswahili